Darshan Ease

Introduction:

Darshan Ticket Booking App is your gateway to seamless travel experiences. Designed to cater to your travel needs with ease and efficiency, Darshan offers a convenient platform to book tickets for various modes of transportation, including buses, trains, and flights. With user-friendly features and a robust booking system, Darshan ensures a hassle-free booking process, allowing you to focus on enjoying your journey. Whether you're planning a solo adventure or a family vacation, Darshan Ticket Booking App is your trusted companion for a smooth and enjoyable travel experience.

DarshanEase refers to the process of reserving slots or purchasing tickets for darshan at various temples. It has become an integral part of the spiritual journey, offering convenience and flexibility to devotees. With the adoption of technology, temple darshan ticket booking has transitioned from traditional methods, such as queuing up or purchasing tickets on-site, to online platforms and mobile applications.

The ease of booking darshan tickets online has transformed the way devotees plan their visits in advance and secure their preferred time slots. Online ticket booking platforms for temple darshan typically provide a user-friendly interface where devotees can browse through available time slots, select specific pooja types, and complete the booking seamlessly from the comfort of their homes or on-the-go.

Team Members:

G. Radhika-Team lead

Ch.Uma Sai Sindhu-Team member

Ch.Yaswanth-Team member

N. Sai Kumar-Team member

V.Rajashekar Reddy

Scenario Based Case Study:

Shiva is a devout individual who finds solace and peace in visiting temples regularly. His spiritual journey involves visiting multiple temples and taking darshans, which are integral to his religious practice and personal beliefs. However, Shiva often faces challenges in planning these visits, including long queues, uncertainty about darshan timings, and difficulty in finding information about multiple temples in one place.

- **User Registration and Authentication:** Enable devotees to register accounts, log in securely, and authenticate their identity to access the temple darshan ticket booking system.
- **Darshan Listings:** Display a comprehensive list of available darshan slots with details such as temple name, location, darshan date, and timings.
- **Temple Selection:** Provide a selection of temples offering darshan slots, including detailed information about each temple's location and unique darshan offerings.
- **Slot Selection:** Allow devotees to choose their preferred darshan slots from the available options, showcasing the layout of the temple and available slots.
- **Ticket Booking:** Empower devotees to book tickets for a specific darshan slot, specifying the number of devotees accompanying them.
- Donation Integration: Integrate with a secure payment gateway to facilitate online donations and contributions along with the darshan ticket booking.
- **Booking Confirmation:** Provide devotees with a confirmation page or email containing the details of the darshan booking, including darshan slot information and booking ID.
- **Booking History:** Allow devotees to view their past and upcoming darshan bookings, offering options to cancel or modify bookings if allowed by the temple.
- **Darshan Slot Availability Tracking:** Keep track of the availability of darshan slots for each temple and update in real-time as slots are booked or canceled.

SYSTEM REQUIREMENTS :

To ensure smooth development, deployment, and usage of the DarshanEase Web Application, certain system prerequisites must be met. These requirements are categorized into software, setup, and hardware specifications, all of which contribute to building a robust and scalable darshan Ease platform.

1. Software Requirements

These are the essential tools and platforms required to develop, test, and run the application efficiently.

- **Operating System**: Windows 10/11, macOS, or Linux — Supports cross-platform development and testing.
- **Node.js (v16 or above)**: Provides the runtime environment for building and managing frontend logic and Powers the server-side logic and handles API routing. 👉 Download Node.js
- **npm (v8 or above)**: A package manager required to install dependencies for React and related libraries.
- **React.js**: A JavaScript library for building dynamic and responsive user interfaces.
- **Browser**: Google Chrome / Firefox (latest version) — For rendering and testing the UI in real-time.
- **Express.js**: A lightweight web framework for building RESTful APIs.
- **MongoDB**: A NoSQL database used to store structured and unstructured data related to users,Admin, Organizers 👉 Download MongoDB
- **Postman**: Tool for testing APIs during development.
- **Visual Studio Code**: Preferred code editor with built-in Git and terminal support.
- **Git & GitHub**: For version control and collaborative development.

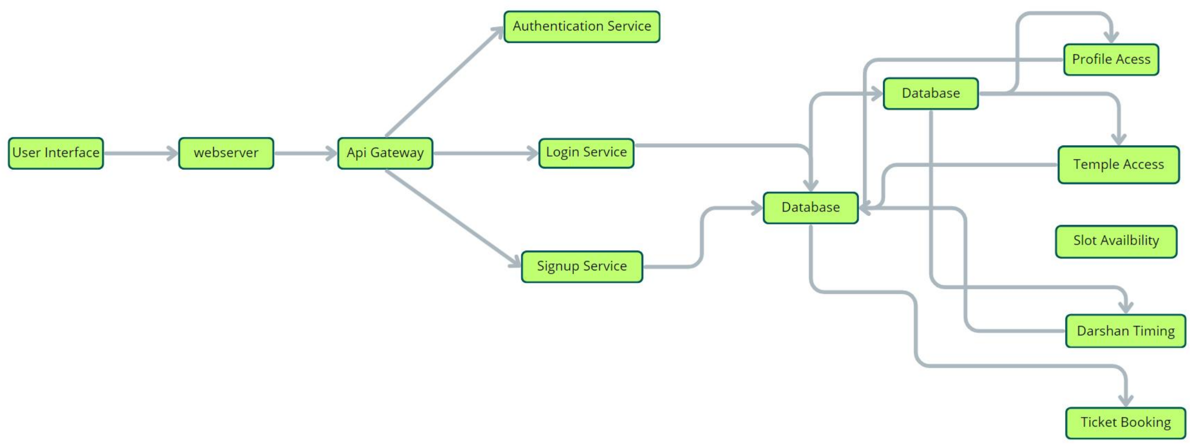
2. Hardware Requirements

Describes the minimum and recommended specifications needed to support the development and usage of the application.

- **Processor**: Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or better — Ensures fast compilation and multitasking during development.
- **RAM**: Minimum 8 GB (16 GB recommended) — For handling development servers, IDEs, and browser testing simultaneously.
- **Storage**: At least 1 GB free space — Required for package installations, MongoDB setup, and local project files.
- **Display**: 1366x768 or higher — Recommended for optimal coding experience and application layout visualization.

PROJECT ARCHITECTURE

Technical Architecture:

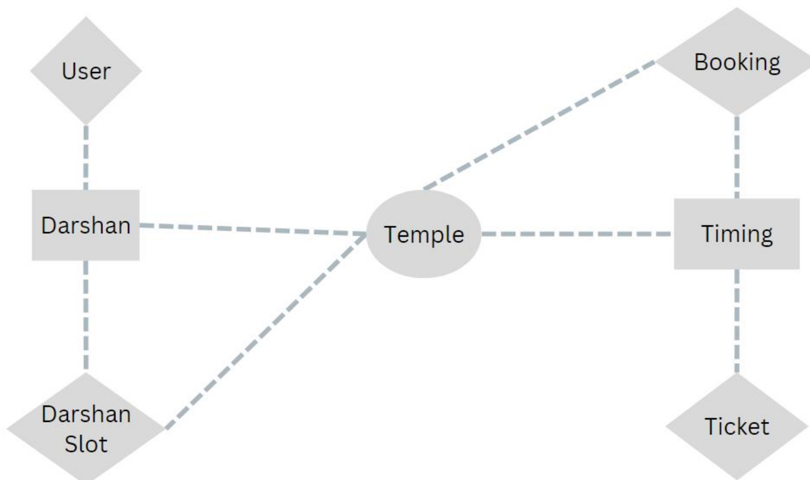


In this technical architecture, the DarshanEase consists of several components:

- User Interface: Represents the web interface through which devotees interact with the system to explore darshan timings, select preferred slots, and make bookings.
- Web Server: Hosts the user interface and serves the web pages to the users.
- API Gateway: Acts as a single entry point for client requests and routes them to the respective services.
- Authentication Service: Handles user authentication and authorization for secure access to the system.
- Database: Stores persistent data related to darshan slots, bookings, payments, and other entities crucial to the temple darshan ticket booking system.
- Darshan Service: Manages temple-related information, including darshan timings, slot availability, and temple access details.
- Booking Service: Handles the process of creating and managing darshan bookings, allowing devotees to secure slots for their temple visits.
- Temple Service: Manages information about temples, including their locations, darshan amenities, and access details.
- Seat Service: Manages seat availability, selection, and reservation for devotees attending temple darshans.

- Ticket Service: Generates and manages electronic tickets for devotees, providing a digital record of their darshan bookings.

ER-Diagram:



User:
UserID (PK)
Name
Email
Phone
Address

Temple:
TempleID (PK)
TempleName
Location
DarshanStartTime
DarshanEndTime

DarshanSlot:
SlotID (PK)
TempleID (FK)
Date
StartTime
EndTime
AvailableSeats

Price

Booking:
BookingID (PK)
UserID (FK)
SlotID (FK)
BookingDate
TotalAmount

The relationships between the entities are as follows:

- **User-Booking Relationship:** One-to-Many relationship between User and Booking (One user can make multiple bookings)
  Foreign Key: UserID in Booking
- **Temple-DarshanSlot Relationship:** One-to-Many relationship between Temple and DarshanSlot (One temple can have multiple darshan slots)
  Foreign Key: TempleID in DarshanSlot
- **Booking-DarshanSlot Relationship:** One-to-Many relationship between DarshanSlot and Booking (One darshan slot can have multiple bookings)
  Foreign Keys: SlotID in Booking, TempleID in DarshanSlot

The foreign key relationships are established between the entities using the primary keys from the respective tables.
Please note that this is just an example, and you can modify or expand it based on your specific requirements for the DarshanEase.

Key Features:

- **Devotee Registration and Authentication:** Enable devotees to register accounts, log in securely, and authenticate their identity to access the temple darshan ticket booking system.
- **Darshan Listings:** Display a comprehensive list of available darshan slots with details such as temple name, location, darshan date, and timings.
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- **Darshan Slot Availability Tracking:** Keep track of the availability of darshan slots for each temple and update in real-time as slots are booked or canceled.
- **•Organizer Dashboard:** Provide temple Registration with an interface to manage temple, darshan slots, devotees, availability, and booking-related activities.
- **Darshan Slot Management:** Organizer is responsible ticket issueing and slot booking.
- **Admin Dashboard:** Provide temple administrators with an interface to manage users, organizers,temples, darshan slots, devotees, availability, and booking-related activities.
- **Reporting and Analytics:** Generate reports and analytics on darshan bookings, popular temples, devotee demographics, and other relevant metrics to gain insights into the temple's activities.
- **Integration with External APIs:** Integrate with third-party APIs for services like temple information, darshan slot updates, or real-time availability for seamless data synchronization.These are just a few key features, and you can customize and expand them based on your specific requirements and the scale of the ticket booking system you are building.

Roles and Responsibility

User:-

- Registration: Devotees are responsible for creating an account on the Temple Darshan Ticket Booking App by providing necessary details like name, email, and password.
- Profile Management: Devotees have access to manipulate their profiles, allowing them to update information such as email, name, and password.
- Darshan Booking: Devotees can view available darshan slots, book specific slots, and acquire electronic tickets for their temple visits.
- Feedback: Provide feedback and ratings for the darshan experience.

- Logout: Finally, they can logout from the DarshanEase.

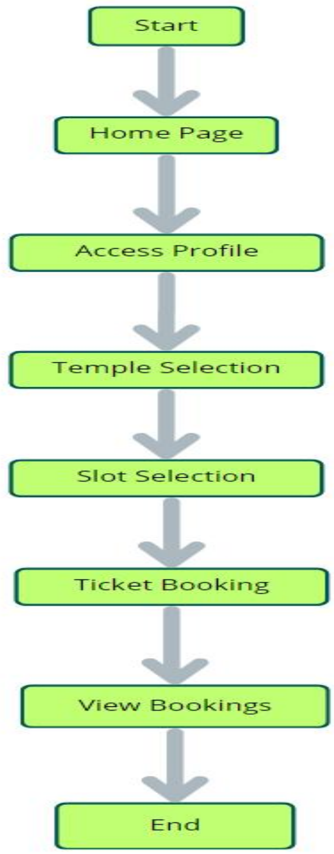
Organizer:-

- Profile Management: Temple organizers have access to manipulate their profiles, allowing them to update information such as email, name, and password.
- Darshan Slot Management: Temple organizers are responsible for managing darshan slots, including scheduling, updating availability, and marking special slots.
- Booking Management: Temple organizers can view and manage darshan bookings, assign slots, and make adjustments as necessary.
- Notification Handling: Receive notifications related to darshan bookings and updates for efficient communication with devotees.
- Logout: Finally, they can logout from the DarshaEase.

Admin:-

- System Management: Admins have full control over all aspects of the temple darshan ticket booking system, overseeing functionalities, configurations, and security.
- Devotee Management: Admins can manipulate devotee information, including creating, updating, and deleting accounts. They also have the authority to manage darshan ratings.
- Temple Organizer Management: Admins can manipulate temple organizer information, including creating, updating, and deleting accounts.
- Facility Management: Admins can schedule darshan slots, allocate resources efficiently, and handle maintenance requests for the temple.
- Events Management: Admins have the authority to create and manage temple events, including scheduling and deleting events.
- Logout: Finally, they can logout from the DarshaEase.

User Flow:

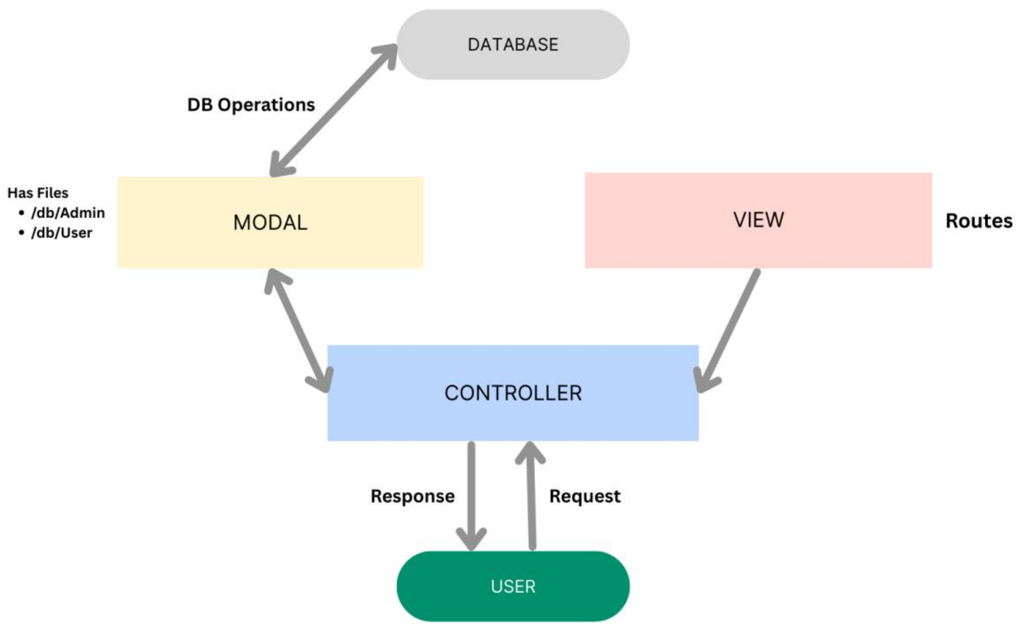


1. Start: The process begins as users enter the Temple Darshan Ticket Booking App.
2. Home Page: Users land on the home page, providing an overview of the temple darshan ticket booking services. From here, they can initiate various actions.
3. Access Profile: Users have the option to access their profiles, allowing them to view or update personal information, preferences, and past booking history.
4. Temple Selection:After accessing their profiles, users proceed to select a specific temple for darshan. The app presents a list of available temples, along with relevant details.
5. Slot Selection: Users navigate through the available darshan slots for the selected temple. They can view darshan timings, slot availability, and choose a preferred time slot for their visit.
6. View Bookings: Users have the option to view their current and past darshan bookings. This section provides details about booked slots, confirmation status, and electronic tickets.
7. Ticket Booking: For new bookings, users can initiate the ticket booking process. This involves selecting a temple, choosing a darshan slot, confirming the booking, and receiving an electronic ticket.

8. End: The flow concludes as users have completed their desired actions within the

This flowchart outlines the user journey within the app, starting from the home page, navigating through profile and temple selections, managing bookings, and concluding with the booking process or other activities, ultimately providing a seamless experience for DarshanEase.
Feel free to modify the code as per your specific requirements or add additional nodes as needed.

MVC Pattern :



The Darshan Ease backend application follows the **Model-View-Controller (MVC)** architectural pattern, a software design approach that separates an application into three interconnected layers. This separation allows for modularity, easier maintenance, and scalability.

Model Layer (Data Layer)

The **Model** layer is responsible for handling all data-related logic. This includes the definition of data schemas and the operations performed on the database using those schemas. The models are implemented using **Mongoose**, which provides a schema-based solution to model application data for MongoDB.

Controller Layer

The **Controller** layer acts as an intermediary between the view (routes) and the model. It receives incoming requests, processes the input (which may include validation or transformation), calls the appropriate methods from the model, and then returns a response to the client.

View Layer (Routing Layer)

In the context of a backend REST API, the **View** is implemented as the **routing layer**, where various endpoints are defined. These endpoints determine how the backend responds to different HTTP requests (GET, POST, PUT, DELETE) and are responsible for invoking the appropriate controller functions.

Advantages of Using MVC in This Project

- **Separation of Concerns**: Each layer has a clearly defined responsibility, improving readability and maintainability.
- **Scalability**: New features can be added easily by creating new routes, controllers, and models.
- **Reusability**: Logic in controllers and models can be reused across multiple parts of the application.
- **Testing**: Each layer can be tested independently, especially the controllers and models.
- **Collaboration-Friendly**: Multiple developers can work simultaneously on different layers without conflict.

PROJECT SETUP AND CONFIGURATION :

- ## Creating project folder

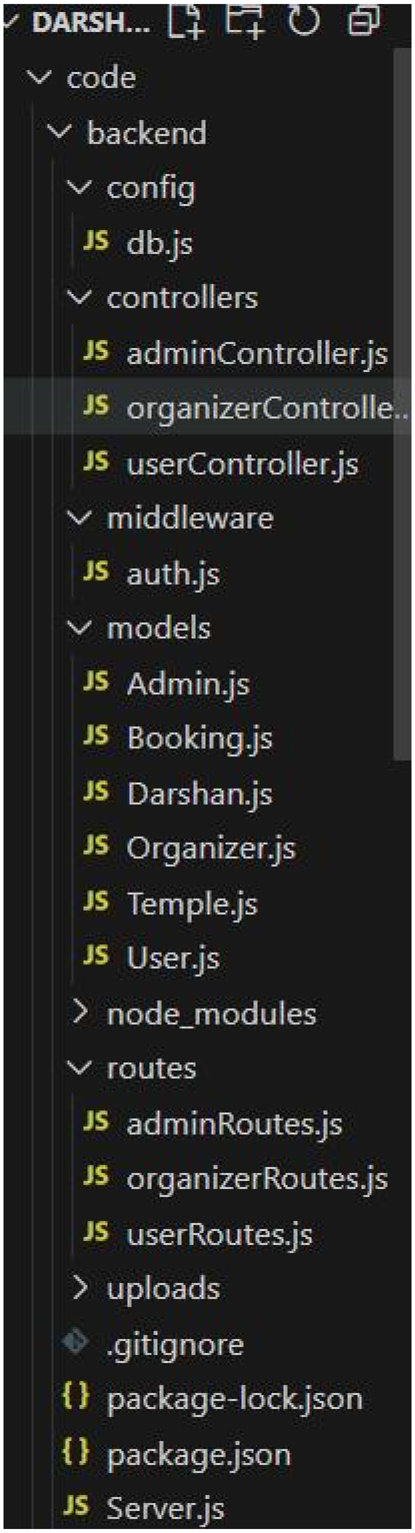
    1. Create a new folder with your DarshanEase.
    2. Inside that folder create two new folders.
    3. Name one as **Client**.
    4. Name another one as **Server**.
    5. Now open that folder in VS Code.

- ## Client setup (installing react app)
    - Open the Client folder in the terminal of VScode.
        - npm create vite@latest . -- --template react
    - Select React framework from the given options.
        - Select a framework:

        - │ React
  - Select JavaScript variant from the given options.
    - Select a variant:
    - │ JavaScript
  - Now lets navigate to the client folder by giving the following command.
    - cd client
  - To install all the packages run the following command.
    - npm install
  - To start the React server type the following command.
    - npm run dev

- ## Server setup (npm init)
  - Open Server folder in terminal of VScode.
    - npm init -y
  - Create files:
    - server.js
  - Create folders:
    - models
    - controllers
    - routes

BACKEND DEVELOPMENT:


DARSH...
code
backend
config
db.js
controllers
adminController.js
organizerControlle...
userController.js
middleware
auth.js
models
Admin.js
Booking.js
Darshan.js
Organizer.js
Temple.js
User.js
node_modules
routes
adminRoutes.js
organizerRoutes.js
userRoutes.js
uploads
.gitignore
package-lock.json
package.json
Server.js

- **Setup express server**
    1. Create index.js file in the server (backend folder).
    2. Create a .env file and define port number to access it globally.
    3. Configure the server by adding cors, body-parser.

- **User Authentication:**
    - Create routes and middleware for user registration, login, and logout.
    - Set up authentication middleware to protect routes that require user authentication.

- **Define API Routes:**
    - Create separate route files for different API functionalities such as users orders, and authentication.
    - Define the necessary routes for listing products, handling user registration and login,managing orders, etc.
    - Implement route handlers using Express.js to handle requests and interact with the database.

- **Implement Data Models:**
    - Define Mongoose schemas for the different data entities like products, users, and orders.
    - Create corresponding Mongoose models to interact with the MongoDB database.
    - Implement CRUD operations (Create, Read, Update, Delete) for each model to perform database operations.

- **User Authentication:**
    - Create routes and middleware for user registration, login, and logout.
    - Set up authentication middleware to protect routes that require user authentication.

- **Error Handling:**
    - Implement error handling middleware to catch and handle any errors that occur during the API requests.
    - Return appropriate error responses with relevant error messages and HTTP status codes.

Database Development :

1. Configure MongoDB:

- Install Mongoose.
- Create database connection.
- Create Schemas & Models.

2. Connect database to backend:

Now, make sure the database is connected before performing any of the actions through the backend. The connection code looks similar to the one provided below.

```javascript
const mongoose = require('mongoose');

mongoose.connect("mongodb://127.0.0.1:27017/yourdbname", {
    useNewUrlParser: true,
    useUnifiedTopology: true
})
.then(() => console.log("✅ Database connected"))
.catch((err) => console.log("❌ DB Connection Failed:", err));
```

3. Configure Schema:

Firstly, configure the Schemas for MongoDB database, to store the data in such pattern. Use the data from the ER diagrams to create the schemas.
The schemas are looks like for the Application.

User.js Admin.js Organizer.js Darshan.js

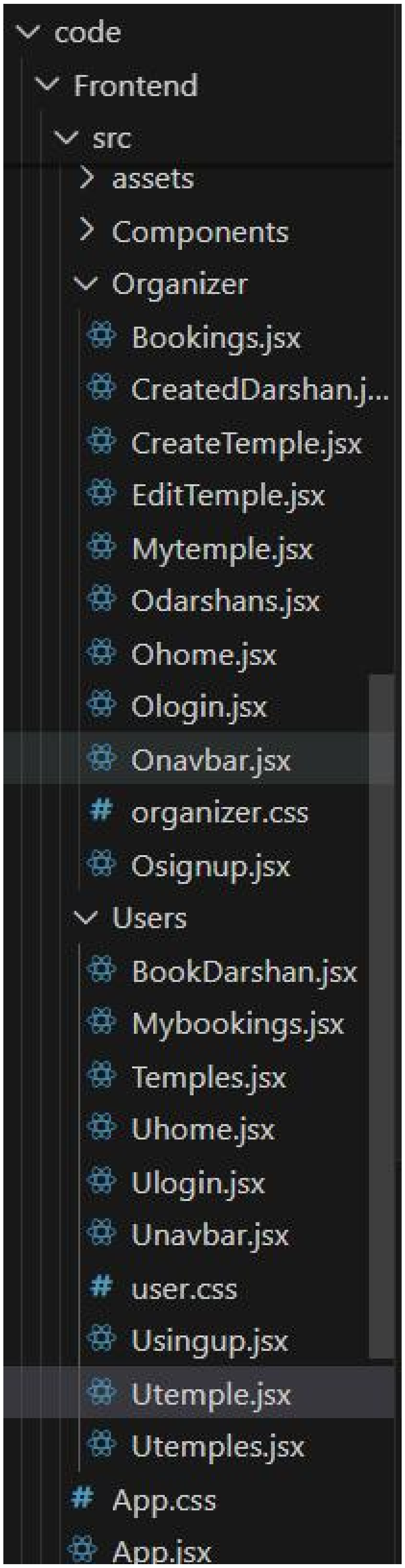
code > backend > models > User.js > ...

```javascript
const mongoose = require('mongoose');

const UserSchema = new mongoose.Schema({
  name: String,
  email: { type: String, unique: true },
  password: String,
  userId: {
    type: mongoose.Schema.Types.ObjectId,
    ref: "User"
  }
});

module.exports = mongoose.model('User', UserSchema);
```

Front-End Development.


code
Frontend
src
assets
Components
Organizer
Bookings.jsx
CreatedDarshan.j...
CreateTemple.jsx
EditTemple.jsx
Mytemple.jsx
Odarshans.jsx
Ohome.jsx
Ologin.jsx
Onavbar.jsx
organizer.css
Osignup.jsx
Users
BookDarshan.jsx
Mybookings.jsx
Temples.jsx
Uhome.jsx
Ulogin.jsx
Unavbar.jsx
user.css
Usingup.jsx
Utemple.jsx
Utemples.jsx
App.css
App.jsx

1. Setup React Application:

• Create React application.

• Configure Routing.

• Install required libraries.

2. Design UI components:

• Create Components.

• Implement layout and styling.

• Add navigation.

3. Implement frontend logic:

• Integration with API endpoints.

• Implement data binding.

Project Execution

- ### Steps for Project execution

Step 1: Set Up the Frontend (React App):

a) Open a terminal and navigate into the client folder:

cd client

b) Once installation is done, start the React development server:
npm run dev

c) The app should now be running on:
http://localhost:5173

Step 2: Set Up the Backend (Express Server)

a) Open a new terminal tab/window or split the terminal.

b) Navigate into the server folder:
cd ../server

Step 3: Configure Environment Variables

a) **Inside the** server f**older, create a new file named .**env **(no file extension).**

b) **In that** .env **file, add your MongoDB connection string:**
MONGO_URI=mongodb://localhost:7000/darshanease

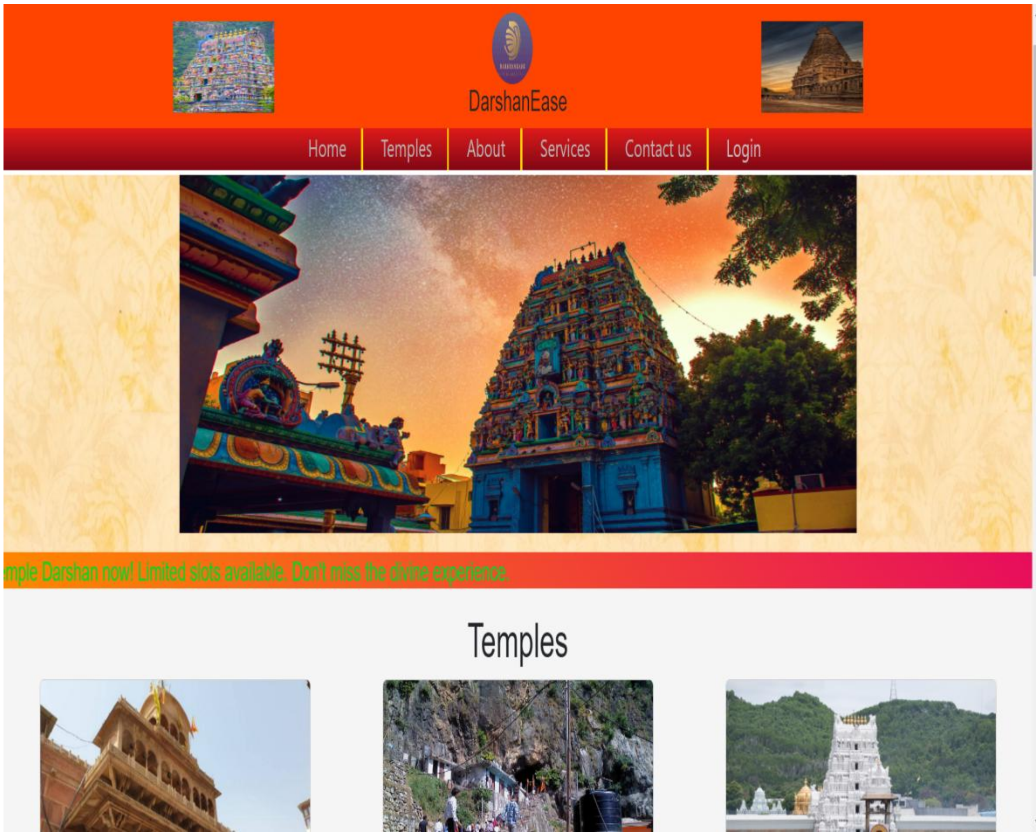
Step 4: Start the Backend Server:

- Inside the same server folder, run the backend server using**:**
  nodemon index.js
- The server should start on:
  http://localhost:8000

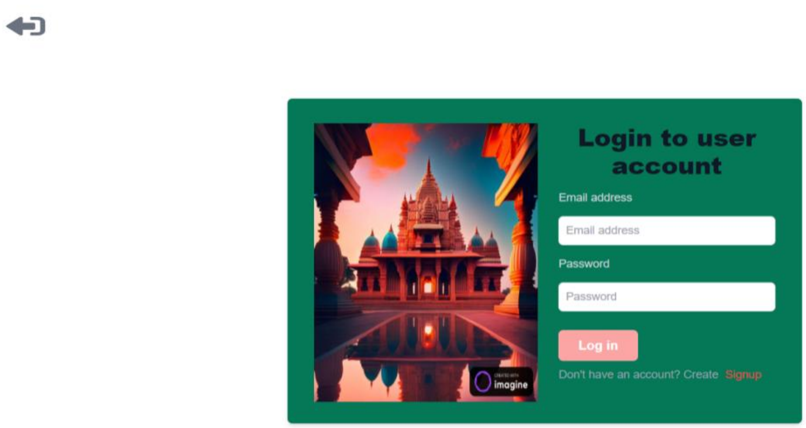
OUTPUT SCREENSHOTS:

Finally, after finishing coding the projects we run the whole project to test it's working process and look for bugs. Now, let's have a final look at the working of our Darshan Ease.

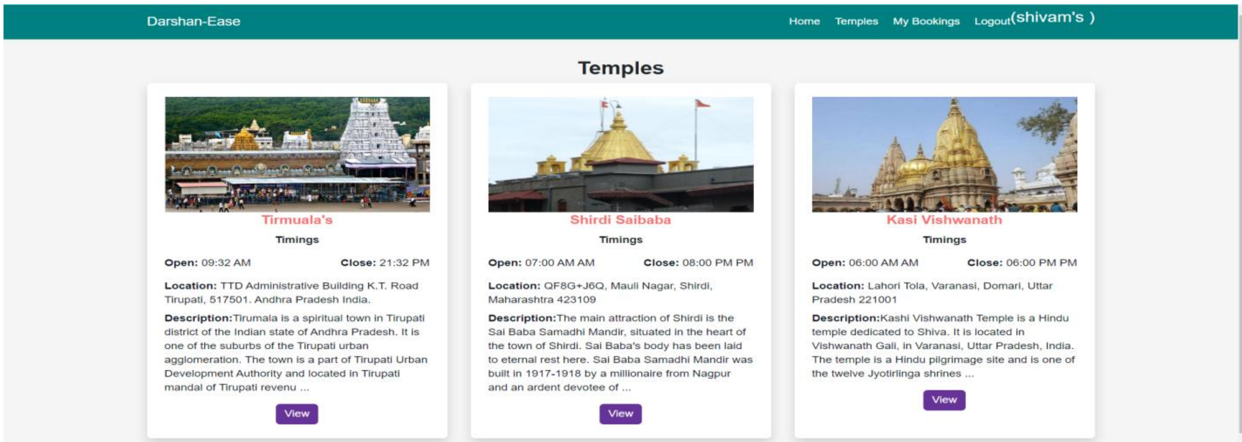
Landing page:-


DarshanEase
Home
Temples
About
Services
Contact us
Login
mple Darshan now! Limited slots available. Don't miss the divine experience.
Temples

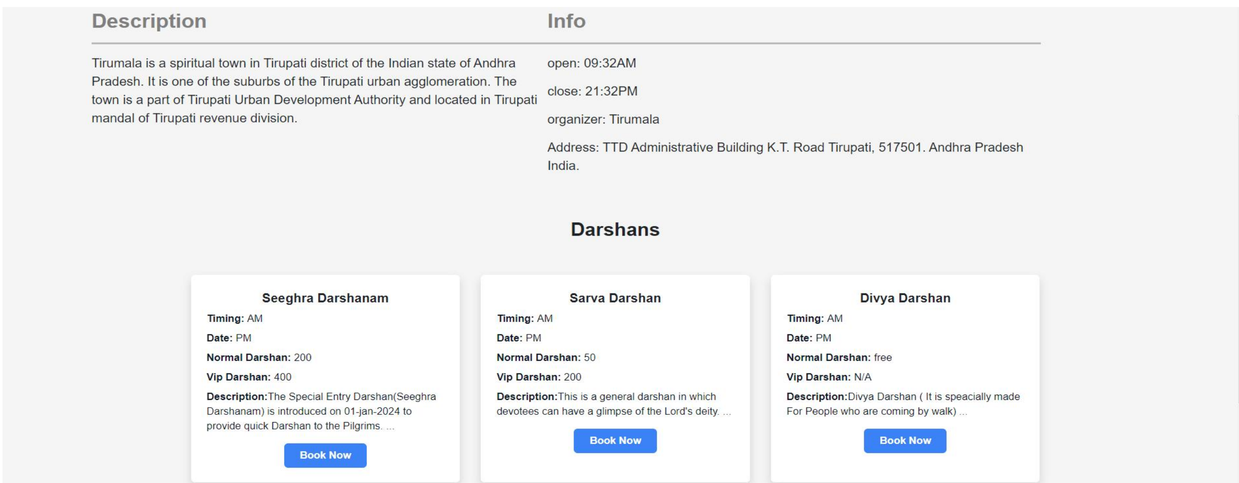
Login Page:-


Login to user account
Email address
Email address
Password
Password
Log in
Don't have an account? Create Signup

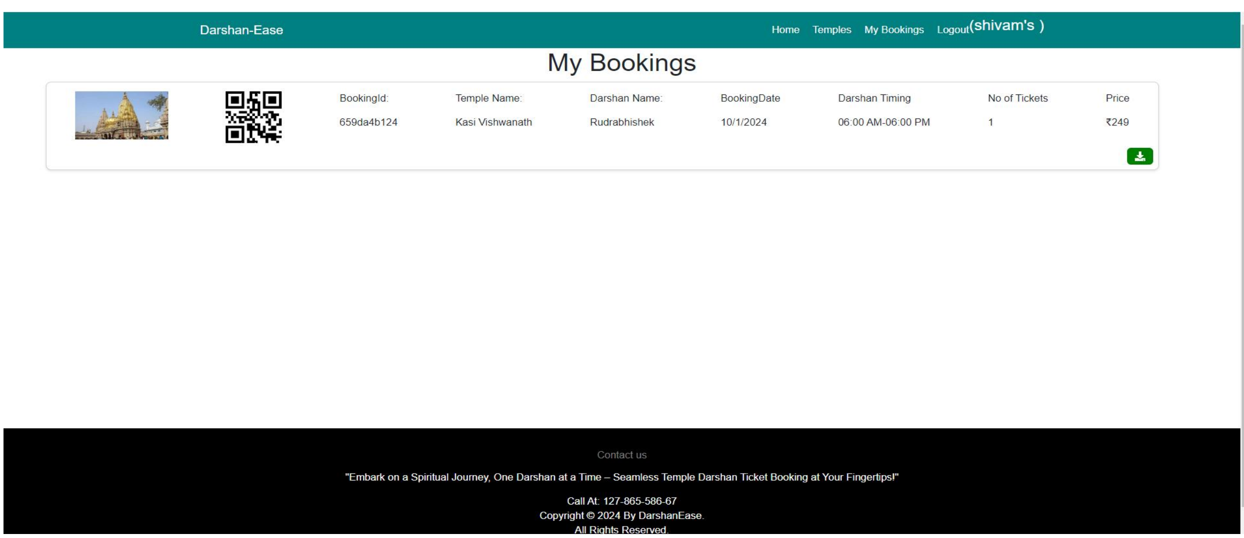
Temples Page:-



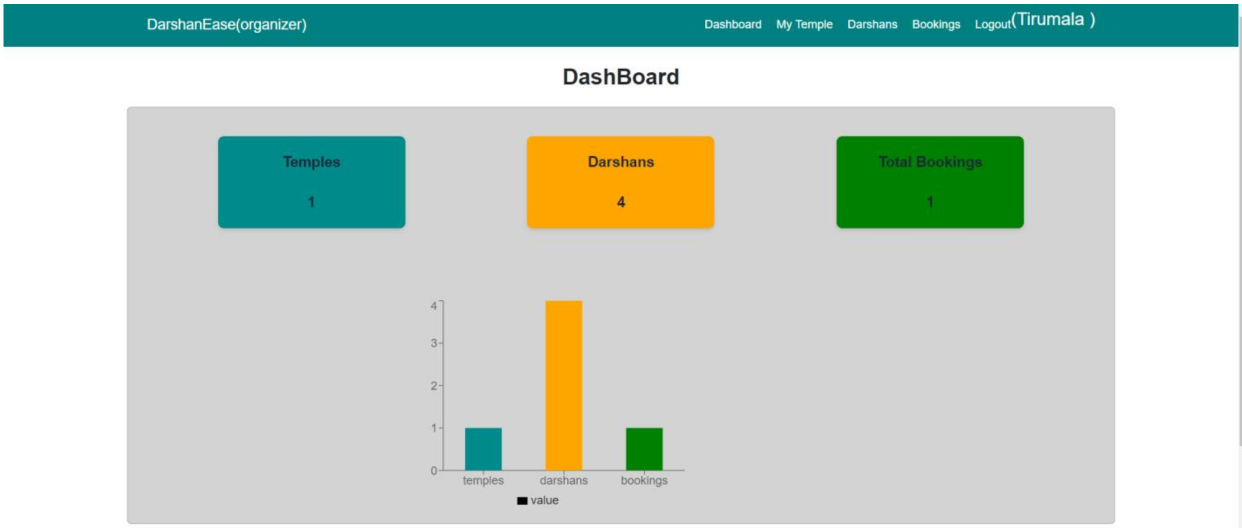
Darshans:-



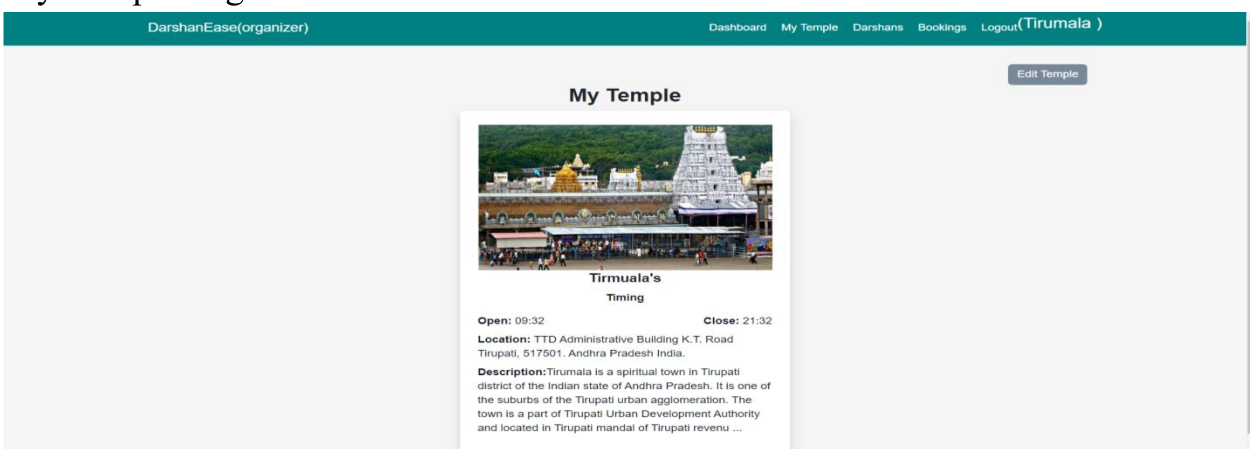
My Bookings Page:-



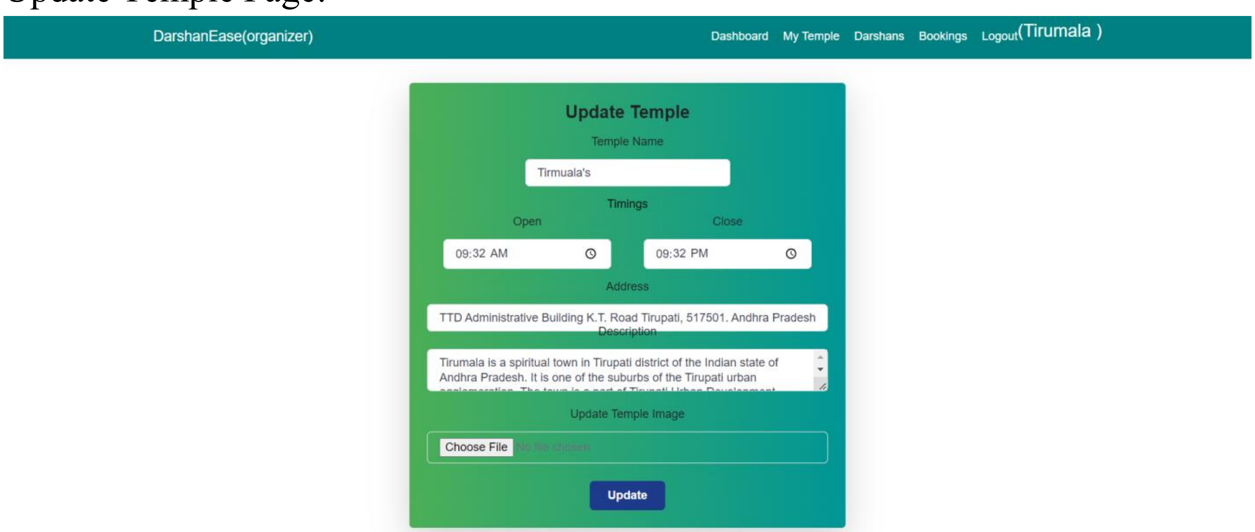
Organizer Dashboard:-



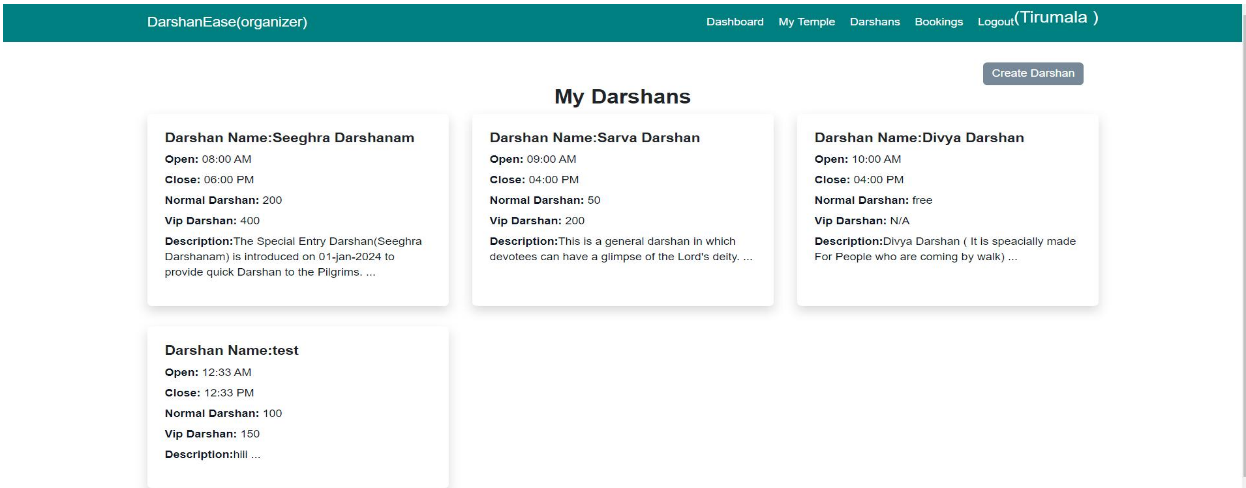
My Temple Page:-



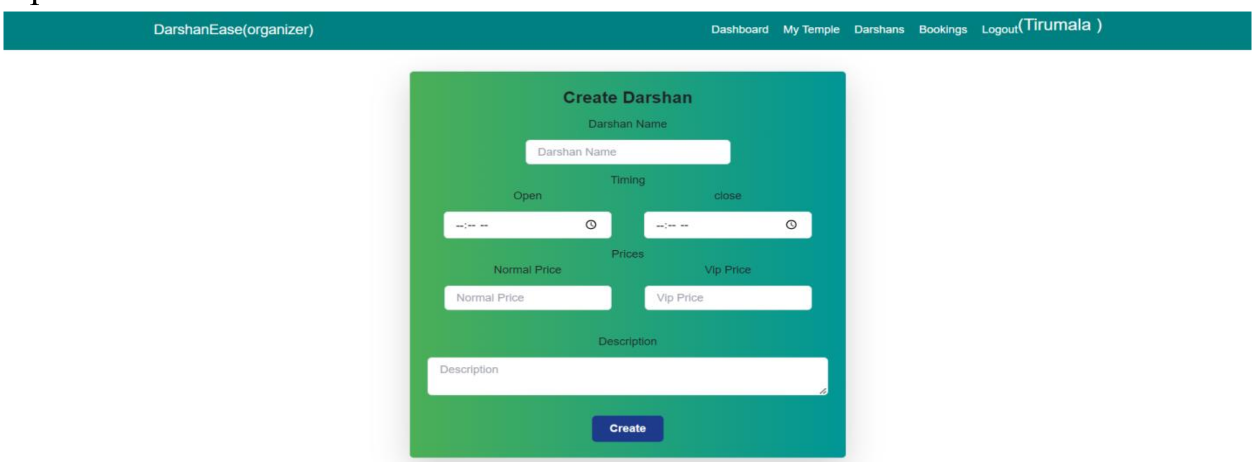
Update Temple Page:



My Darshans Page:-



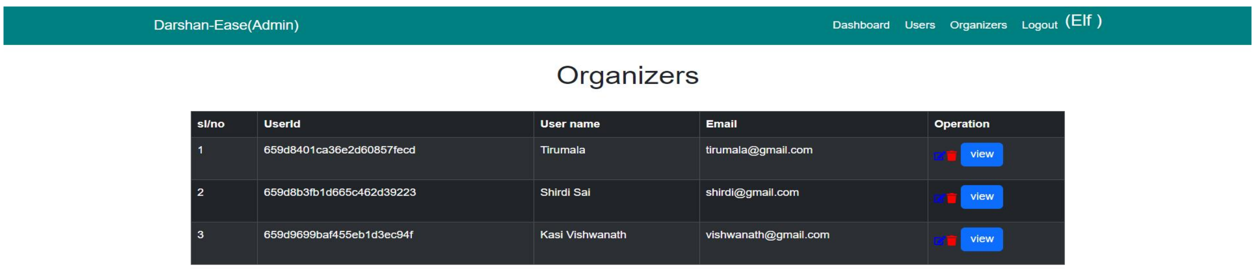
Update Darshan:-

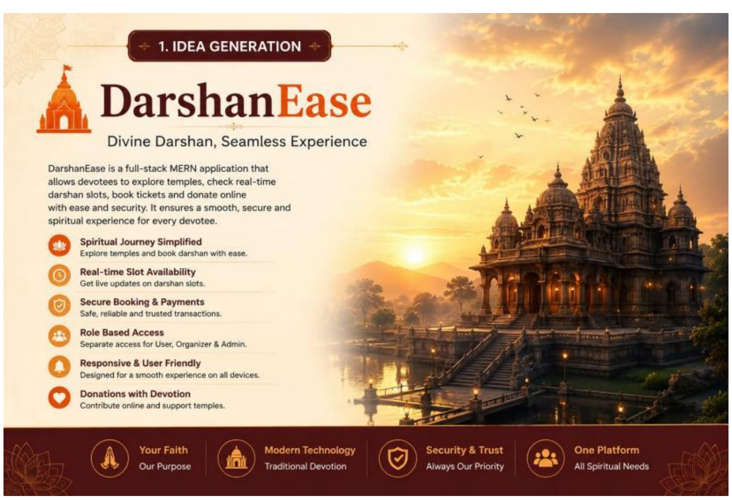


Admin Dashboard Page:-

Users page:

Organizers





Future Enhancements for DarshanEase:

The following features can be added in future versions of the **DarshanEase Temple Darshan Ticket Booking System** to improve functionality and user experience:

1. AI-Based Crowd Prediction

- Analyze historical booking data.
- Predict crowd levels for upcoming days and festivals.
- Suggest less crowded time slots to devotees.

2. Waiting List Management

- Allow devotees to join a waiting list when slots are full.
- Automatically allocate cancelled slots to waiting users.

3. Multi-Language Support

- Provide the application in multiple languages such as:
    - English
    - Telugu
    - Hindi
    - Tamil
    - Kannada

4. QR Code Verification

- Generate QR-based tickets.
- Temple staff can scan QR codes at entry points for quick verification.

5. SMS and Email Notifications

- Send booking confirmations and reminders through:
    - SMS
    - Email
    - Push notifications

6. Mobile Application

- Develop Android and iOS mobile applications for better accessibility.

7. Live Queue Monitoring

- Display real-time queue status and estimated waiting times.
- Help devotees plan their arrival accordingly.

8. Navigation and Route Guidance

- Integrate maps for directions to temples.
- Provide nearby parking and accommodation information.

9. Online Pooja and Seva Booking

- Allow users to book:
    - Archana
    - Abhishekam
    - Special Poojas
    - Festival Sevas

10. Donation and Charity Management

- Enable online donations for:
    - Temple development
    - Annadanam programs
    - Charity activities

11. Facial Recognition Entry System

- Use facial recognition technology for faster and secure entry verification.

12. Advanced Analytics Dashboard

- Provide detailed reports for temple organizers including:
    - Daily visitors
    - Revenue analysis
    - Peak crowd timings
    - Popular darshan slots

13. Festival Event Management

- Manage special events and festival bookings separately.
- Handle large crowds efficiently during festivals.

14. Accessibility Features

- Provide dedicated services for:
    - Senior citizens
    - Differently-abled devotees
    - VIP darshan visitors

15. Blockchain-Based Ticket Verification

- Use blockchain technology to prevent ticket duplication and fraud.

The demo of the app is available at:-

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- Predict crowd levels for upcoming days and festivals.
- Suggest less crowded time slots to devotees.

2. Waiting List Management

- Allow devotees to join a waiting list when slots are full.
- Automatically allocate cancelled slots to waiting users.

3. Multi-Language Support

- Provide the application in multiple languages such as:

- English
- Telugu
- Hindi
- Tamil
- Kannada

4. QR Code Verification

- Generate QR-based tickets.
- Temple staff can scan QR codes at entry points for quick verification.

5. SMS and Email Notifications

- Send booking confirmations and reminders through:
    - SMS
    - Email
    - Push notifications

6. Mobile Application

- Develop Android and iOS mobile applications for better accessibility.

7. Live Queue Monitoring

- Display real-time queue status and estimated waiting times.
- Help devotees plan their arrival accordingly.

8. Navigation and Route Guidance

- Integrate maps for directions to temples.
- Provide nearby parking and accommodation information.

9. Online Pooja and Seva Booking

- Allow users to book:
    - Archana
    - Abhishekam
    - Special Poojas
    - Festival Sevas

10. Donation and Charity Management

- Enable online donations for:
  - Temple development
  - Annadanam programs
  - Charity activities

11. Facial Recognition Entry System

- Use facial recognition technology for faster and secure entry verification.

12. Advanced Analytics Dashboard

- Provide detailed reports for temple organizers including:
  - Daily visitors
  - Revenue analysis
  - Peak crowd timings
  - Popular darshan slots

13. Festival Event Management

- Manage special events and festival bookings separately.
- Handle large crowds efficiently during festivals.

14. Accessibility Features

- Provide dedicated services for:
  - Senior citizens
  - Differently-abled devotees
  - VIP darshan visitors

15. Blockchain-Based Ticket Verification

- Use blockchain technology to prevent ticket duplication and fraud.

Demo video: Future Enhancements for DarshanEase

The following features can be added in future versions of the **DarshanEase Temple Darshan Ticket Booking System** to improve functionality and user experience:

1. AI-Based Crowd Prediction

- Analyze historical booking data.
- Predict crowd levels for upcoming days and festivals.
- Suggest less crowded time slots to devotees.

2. Waiting List Management

- Allow devotees to join a waiting list when slots are full.
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Demo video:

https://drive.google.com/file/d/1cLkgT8GgDoEpMcSCH2Rfis7XdOrZXlWX/view?usp=drivesdk